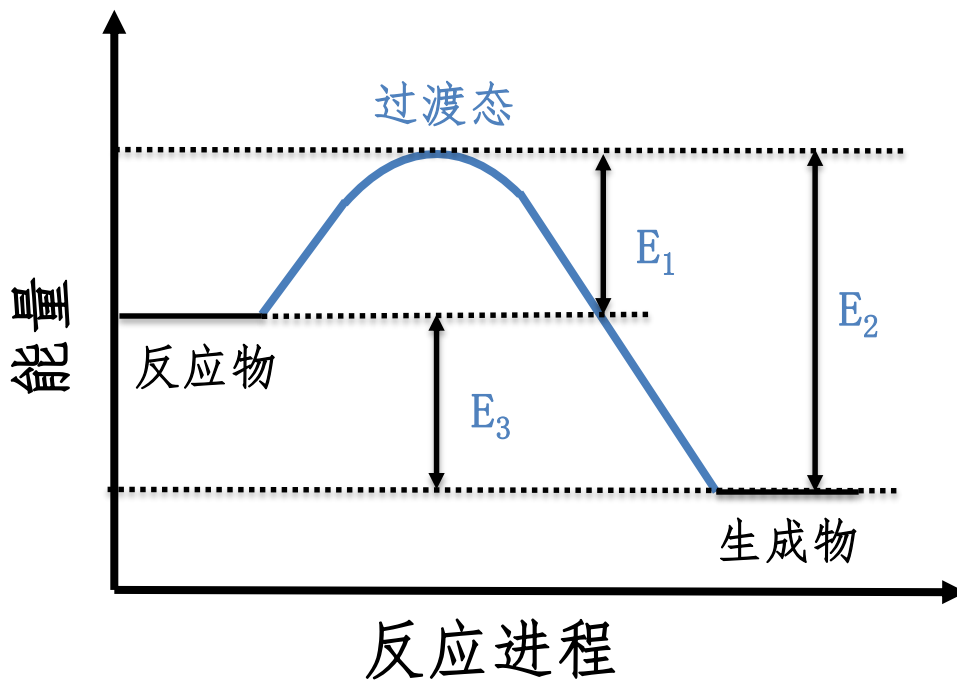


电催化计算实战入门

动力学计算

主讲人：高怡静

过渡态理论是1935年由艾林(Eyring)和波兰尼(Polany)等人提出，过渡态理论建立在统计热力学和量子力学的基础上。

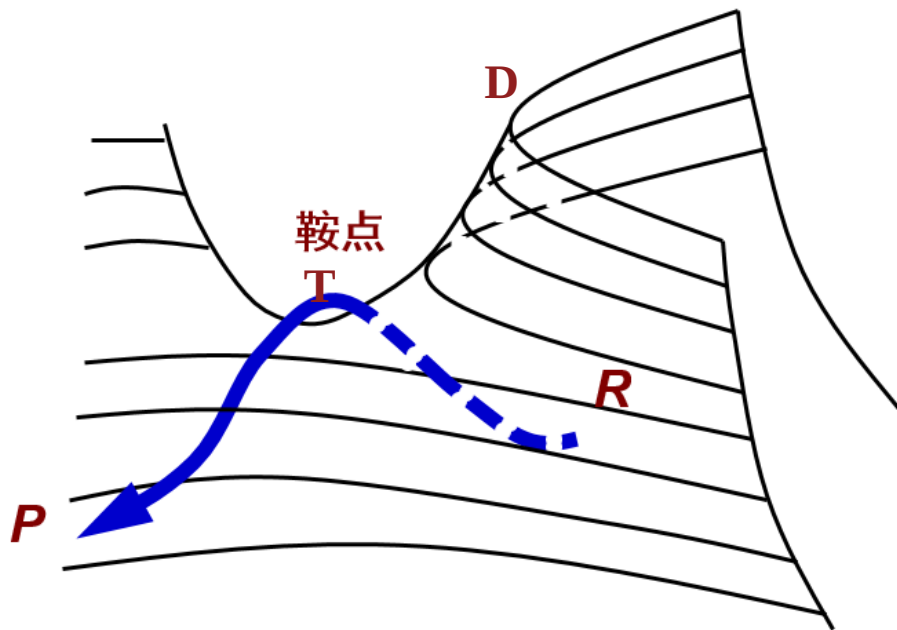


由反应物分子变成生成物分子，中间一定要经过一个**过渡态**，而形成这个过渡态必须吸取一定的**活化能**，这个过渡态就称为活化络合物，所以又称为活化络合物理论。

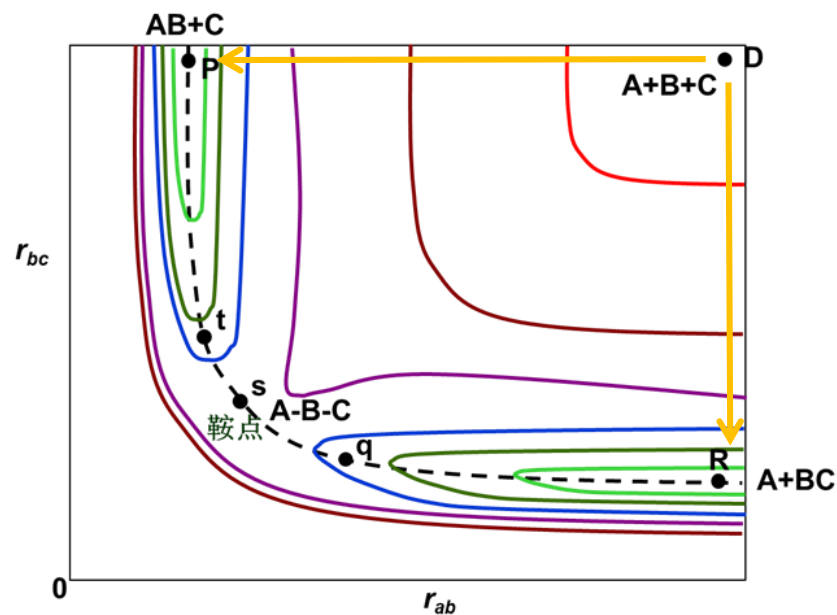
势能面：



立体图：马鞍面



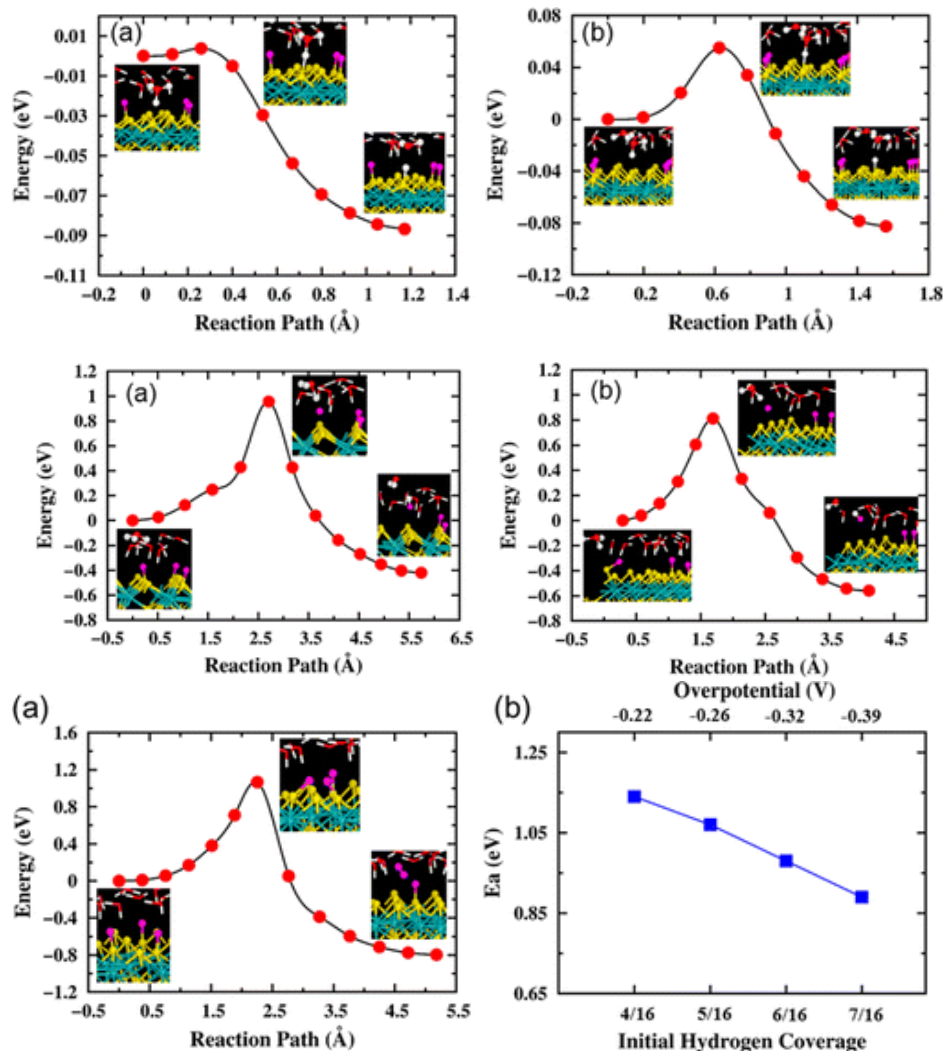
平面图：投影面



鞍点的势能与反应物和生成物所处的稳定态能量R点和P点相比是最高点，但与坐标原点一侧和D点的势能相比是最低点。

D点向P点的过程，势能逐渐降低且P为最低点；D点向R点的过程，势能逐渐降低且R为最低点；所有平行线同理。因此，PsR路径是势能最低的路径。³

能获得的信息



需要准备的文件

/home/cailiaoren1/calss5/NH3/example2

名字

..
 00
 01
 02
 03
 04
 05
 06
 07
 08
 09
 INCAR
 KPOINTS
 log
 POSCAR_fin
 POSCAR_in
 POTCAR
 vasprun.xml

1. 始态结构优化后的
CONTCAR(此处需要存在
POSCAR)和**OUTCAR**

2. 插点得到的中间态结
构的**POSCAR**

3. 末态结构优化后的
CONTCAR(此处需要存在
POSCAR)和**OUTCAR**

4. 其他提交文件

ACS Catal. 2016, 6, 8, 4953–4961

始末态核对和插点方法

准备工作：小脚本的下载

VTST•Tools

VTST

- Download
- Installation
- NEB
- Dynamical Matrix
- Dimer
- Lanczos
- Optimizers
- Scripts
- References

Development

- Source Code Browser
- Bug Tracker

This page

- Download
 - Tar files
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Download

VTST•Tools is released under the [Apache License version 2](#).

Tar files

VTST Code: [vtstcode-184.tgz](#)

VTST Scripts: [vtstscripts.tgz](#)

使用VTST的VASP版本

处理过渡态计算的VTST脚本

SVN Access

A read-only copy of the code can be anonymously checked out of the subversion repository with the following command:

```
svn co http://theory.cm.utexas.edu/svn/vtsttools
```

If you are a developer you can checkout the code using the command:

```
svn co svn+ssh://username@theory.cm.utexas.edu/svn/vtsttools
```

This will fetch a copy of the latest code to a local directory named vtsttools. For more information on subversion read its [online documentation](#).

<http://theory.cm.utexas.edu/vtsttools/download.html>

VTST•Tools

VTST

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SCRIPTS

- General Scripts
- File Conversion Scripts
- Nudged Elastic Band Scripts
- Charge Density Scripts
- Dimer Scripts
- Dynamical Matrix
- Density of State Scripts

Page Source

SCRIPTS

The following are a set of scripts to perform common tasks to help with VASP calculations, and particularly with transition state finding. The included Vasp.pm perl module contains several simple routines that are used by many of the scripts.

Download the scripts: vtstscripts.tgz

Install by uncompressing this file, and adding the vtstscripts directory to your path.

The scripts are organized into the following categories (portals to them are on the left):

- general
- file conversion
- nudged elastic band
- dynamical matrix
- dimer
- charge density
- density of states

Notes:

- We recommend that the first line in the POSCAR file contain the element symbols, in the same order as they appear in the POTCAR. This will allow for proper visualization when files are converted to xyz files.
- For NEB scripts, there needs to be OUTCAR files for the initial and final states placed in the 00 and NI+1, respectively. Please direct questions about these scripts to the discussion [forum](#).

始末态核对和插点方法

dist.pl

usage: dist.pl (POSCAR 1) (POSCAR 2)

output: root sum squared distance between configuration files

计算每个对应原子的初末态距离的平方和，再开根号，来判断两个优化后结构 (CONTCAR) 的相似程度。

nebmake.pl

usage: nebmake.pl (POSCAR1) (POSCAR2) (number of images, NI)

output: directories [00,NI+1] containing initial NEB POSCAR files

Takes initial and final POSCAR files, and linearly interpolates the specified number of images between them. The interpolated files are written to the directories 00 to NI+1, where NI is the number of specified images.

在始末态之间，插入NI个中间态。

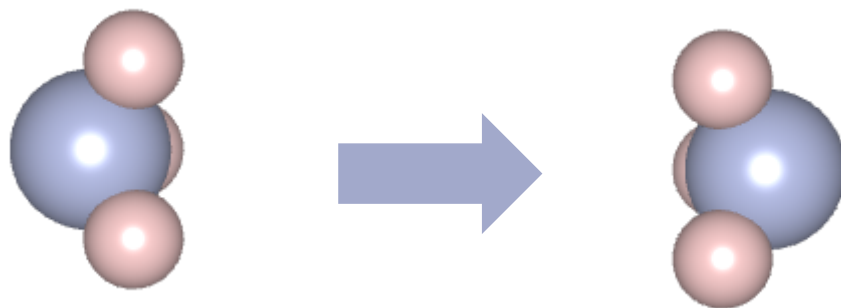
nebef.pl

usage: nebef.pl

output: energy and force of images in the neb

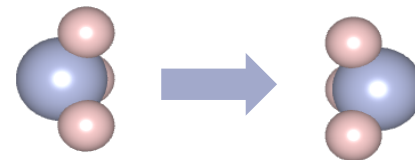
查看neb计算过程中或结束后每个插入点的能量和力收敛情况。

实例：如何给 NH_3 转向过程进行插点



1. 如何核对始末态结构差异
2. 如何进行插点生成POSCAR

NH₃转向过程进行插点



a. 准备始末态两个文件在文件夹里面

 POSCAR_fin	1 KB	2021/10/27 15:27:16	rw-rw-r--	cailiao...
 POSCAR_in	1 KB	2021/10/27 15:27:44	rw-rw-r--	cailiao...

b. 用 **dist.pl** 脚本检查两个优化后结构的相似程度

```
[cailiaoren1@iZuf6ewt8qjkgzelxd4lyr9Z ~]$ cd /home/cailiaoren1/do/5do
INCAR KPOINTS POSCAR_fin POSCAR_in POTCAR
[cailiaoren1@iZuf6ewt8qjkgzelxd4lyr9Z 5do]$ dist.pl POSCAR_in POSCAR_fin
1.36219566848232
```

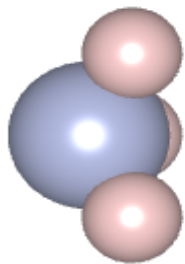
c. nebmake.pl 对始末态进行插点

```
[cailiaoren1@iZuf6ewt8qjkzelxd4lyr9Z 5do]$ nebmake.pl POSCAR_in POSCAR_fin 2  
filetype1: vasp5  
filetype2: vasp5
```

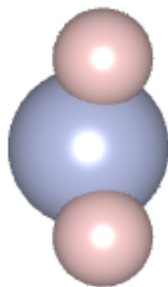
OK, ALL SETUP HERE

FOR LATER ANALYSIS, PUT OUTCARs IN FOLDERS 00 and 03 !!!

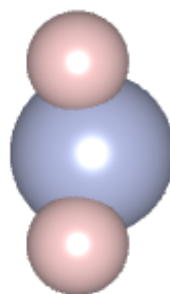
d. 最后检查一下始末态和中间态的结构



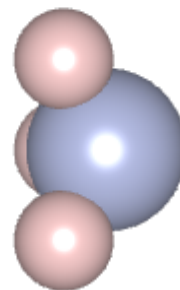
初态 (00)



中间态 (01)



中间态 (02)

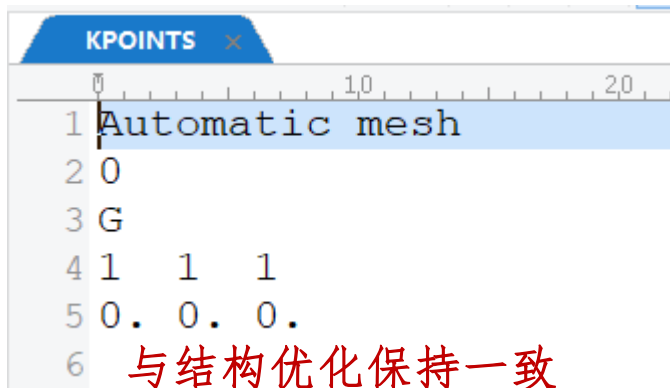


末态 (03)

```
1 SYSTEM = CINEB
2
3 # start paramater for this run
4 ISTART = 0
5 ICHARG = 2
6 PREC= high
7 #ISPIN = 2      !1-nonspin, 2-sp
8
9 #electronic relaxation
10 ENCUT = 500    ! energy cutoff f
11 NELM = 500
12 EDIFF = 1E-04  ! Energy conv
13
14 # ionic relaxation
15 NSW = 600      ! 0 ionic steps
16 IBRION = 3     ! geometry optimiz
17 ISIF = 2       ! optimize geometr
18 EDIFFG = -0.05 ! Force
19
20 # DOS related
```

```
22 ISMEAR =      0  ! Metallic
23 SIGMA =      0.2
24 LWAVE = .FALSE.
25 LCHARG = .FALSE.
26 NPAR = 2
27 GGA=PE
28
29 #VDW
30 IVDW = 11      ! zero damping DFT-D
31
32 #NEB
33 LCLIMB=T       !Ture-CINEB
34 IMAGES = 2      !image numbers
35 SPRING = -5     !Spring force bet
36 POTIM = 0
37 IOPT=1
```

KPOINTS



```
KPOINTS
1 Automatic mesh
2 0
3 G
4 1 1 1
5 0. 0. 0.
6
```

与结构优化保持一致

POTCAR

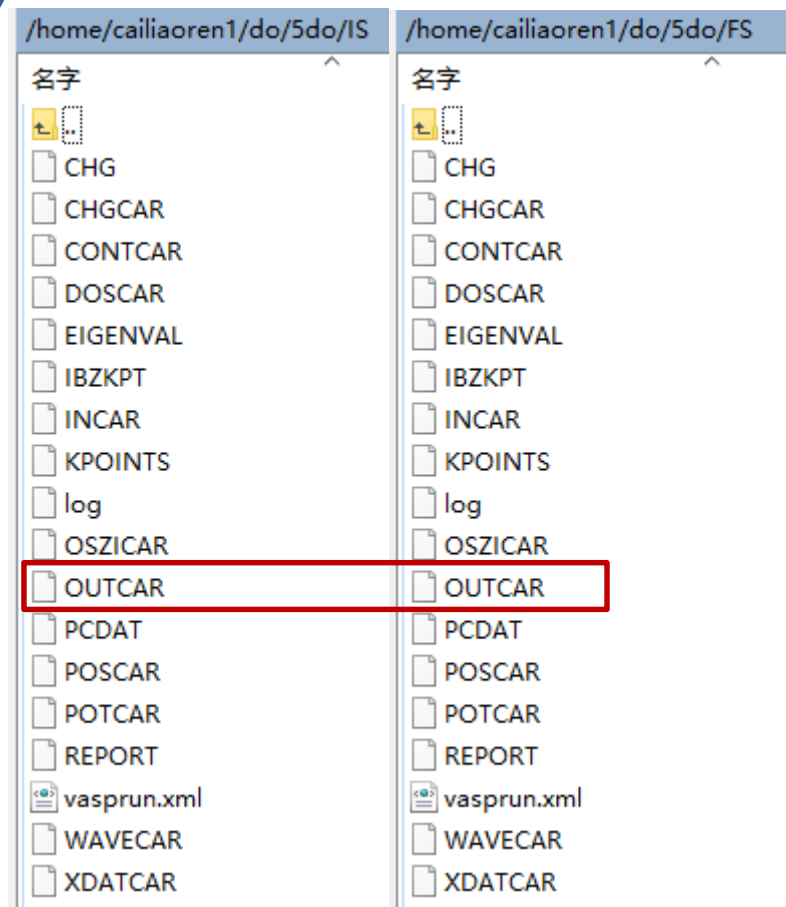
```
[cailiaoren1@iZuf6ewt8qjKzelxd4lyr9Z 5do]$ grep VRHFIN POTCAR
```

```
VRHFIN =H: ultrasoft test
```

```
VRHFIN =N: s2p3
```

与结构优化保持一致

始末态的信息



名字	名字
CHG	CHG
CHGCAR	CHGCAR
CONTCAR	CONTCAR
DOSCAR	DOSCAR
EIGENVAL	EIGENVAL
IBZKPT	IBZKPT
INCAR	INCAR
KPOINTS	KPOINTS
log	log
OSZICAR	OSZICAR
OUTCAR	OUTCAR
PCDAT	PCDAT
POSCAR	POSCAR
POTCAR	POTCAR
REPORT	REPORT
vasprun.xml	vasprun.xml
WAVECAR	WAVECAR
XDATCAR	XDATCAR

为了方便读取，也可以不准备

检查好输入文件和**VASP**的版本之后即可提交。

/home/cailliaoren1/do/5do

名字

..

00

01

02

03

FS

IS

INCAR

KPOINTS

POSCAR_fin

POSCAR_in

POTCAR

```
#!/bin/sh -f
```

```
#PBS -N vasp61
```

```
#PBS -l nodes=1:ppn=2
```

```
#PBS -l walltime=4:00:00
```

```
#PBS -q workq
```

```
#PBS -V
```

```
#PBS -S /bin/bash
```

```
cd $PBS_O_WORKDIR
```

```
##### ENVIRONMENT #####
```

```
source /opt/software/intel2018/bin/compilervars.sh intel64
```

```
NP=`cat $PBS_NODEFILE | wc -l`
```

```
cat $PBS_NODEFILE > hosts
```

```
##### SET EXEC #####
```

```
EXEC=/opt/software/vasp.6.1.0-all/bin/vasp_std
```

```
###EXEC=/opt/software/vasp.6.1.0-all/bin/bin/vasp_ncl
```

```
###EXEC=/opt/software/vasp.6.1.0-all/bin/bin/vasp_gam
```

提交之后多查看。

一看每个结构的优化进度。

```
[cailiaoren1@iZuf6ewt8qjkzelxd4lyr9Z 01]$ grep F OSZICAR
1 F= -.19052941E+02 E0= -.19052941E+02 d E =-.190529E+02
2 F= -.19080794E+02 E0= -.19080794E+02 d E =-.278530E-01
3 F= -.19245353E+02 E0= -.19245353E+02 d E =-.164559E+00
4 F= -.19321765E+02 E0= -.19321765E+02 d E =-.764123E-01
5 F= -.19323508E+02 E0= -.19323508E+02 d E =-.174298E-02
```

二看每个结构间的能量变化趋势。

```
[cailiaoren1@iZuf6ewt8qjkzelxd4lyr9Z example1]$ nebef.pl
0          0.095832          -19.529600          0.000000
1          0.016655          -19.323500          0.206100
2          0.095838          -19.529600          0.000000
```

对比一下插不同点数的结果。

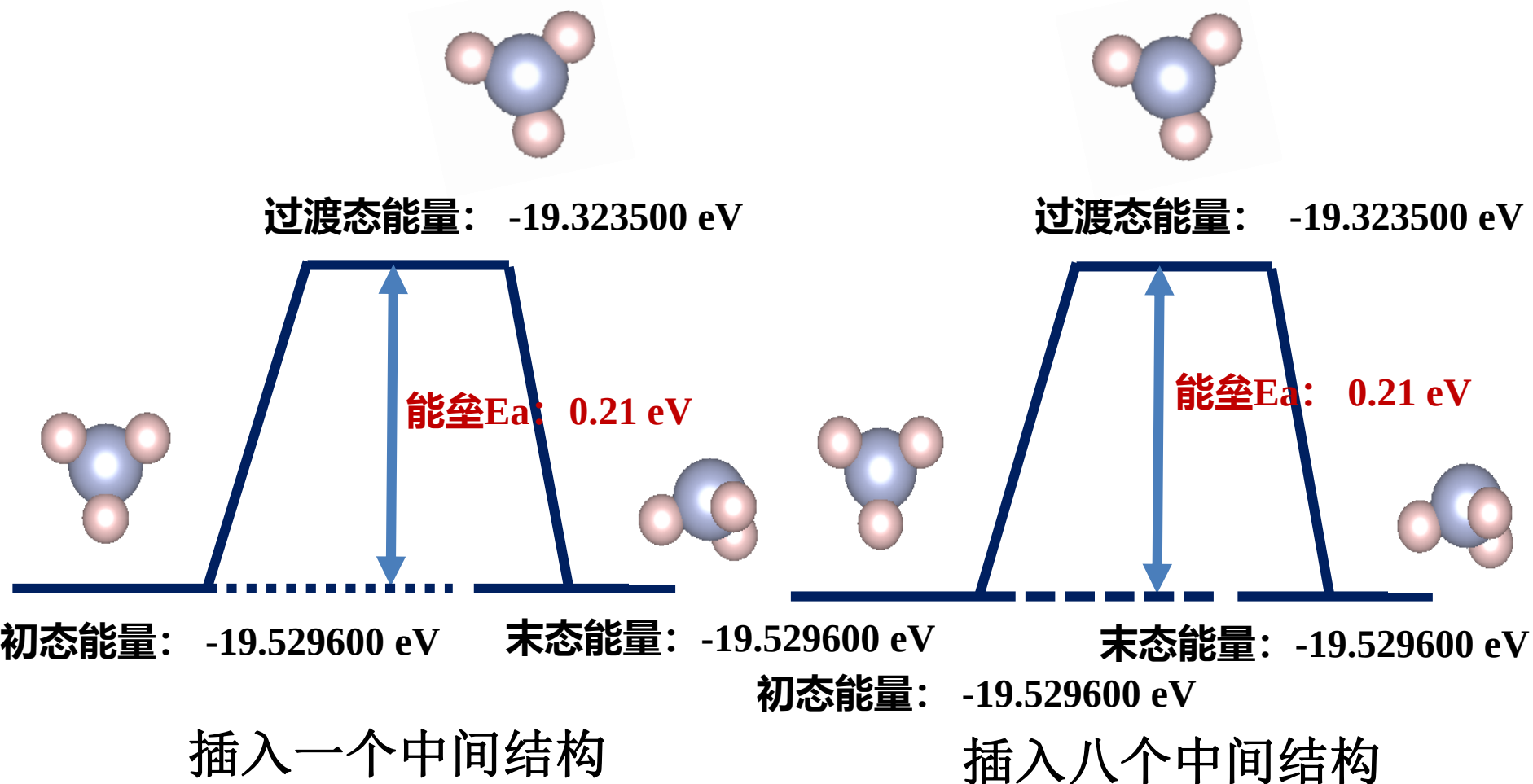
插入一个中间结构

```
[root@iZuf6ewt8qjkd4l9r9Z example1]# nebef.pl
0          0.095832          -19.529600          0.000000
1          0.016655          -19.323500          0.206100
2          0.095838          -19.529600          0.000000
```

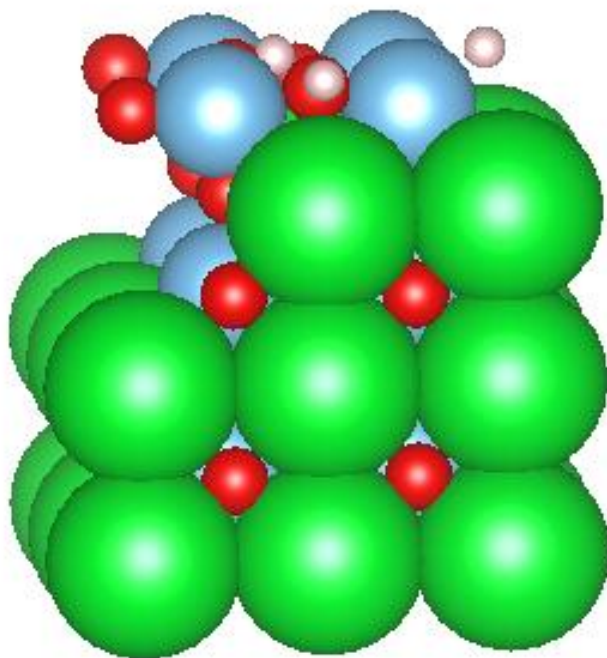
插入八个中间结构

```
[root@iZuf6ewt8qjkd4l9r9Z example2]# nebef.pl
0          0.095832          -19.529600          0.000000
1          0.005114          -19.496500          0.033100
2          0.021105          -19.420800          0.108800
3          0.025934          -19.351900          0.177700
4          0.016976          -19.323500          0.206100
5          0.014575          -19.340300          0.189300
6          0.011312          -19.387800          0.141800
7          0.009416          -19.451300          0.078300
8          0.006417          -19.507900          0.021700
9          0.095838          -19.529600          0.000000
```

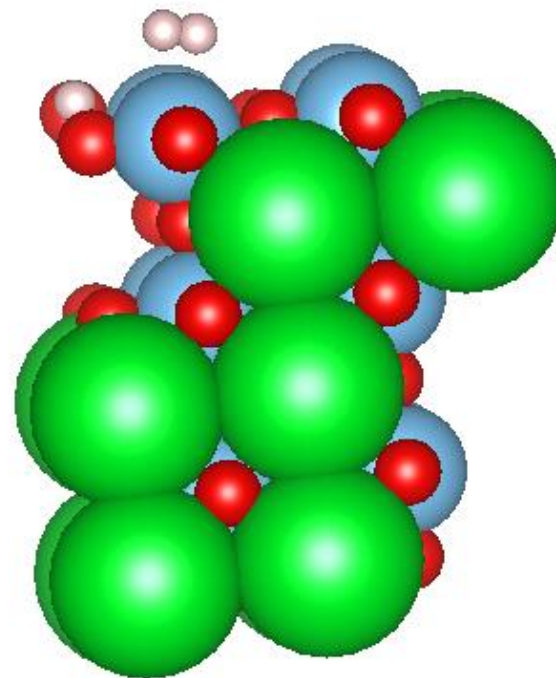
对比一下插不同点数的结果。



基元反应： $*\text{H} + *\text{H} \text{-----} *\text{H}_2$



初态结构



末态结构

初态结构和末态结构优化，优化方法与之前一样

/root/example/class5/test/IS

名字



..



INCAR



KPOINTS



POSCAR



POTCAR



vasp.pbs

Automatic mesh

```
0
G
3 3 1
0. 0. 0.
```

PREC = Normal

ISTART = 0

ICHARG = 2

ISPIN=2

ENCUT = 400

NELM = 500; #maximum of electronic step

EDIFF = 1E-5

LREAL = Auto

EDIFFG = -1E-2

NSW = 1000 # number of ionic step

IBRION = 2 # optimization

ISIF = 2

ISMEAR = 0

SIGMA = 0.2

IVDW=11

VDW_RADIUS=50.2

VDW_CNRRADIUS=20.0

VDW_S6=14.0

VDW_S8=16.0

VDW_SR=1.0

GGA=PE

初态结构优化

初态结构和末态结构优化，优化方法与上次课一样

/root/example/class5/test/FS

名字

..

INCAR

KPOINTS

POSCAR

POTCAR

vasp.pbs

Automatic mesh

```
0
G
3 3 1
0. 0. 0.
```

PREC = Normal

ISTART = 0

ICHARG = 2

ISPIN=2

ENCUT = 400

NELM = 500; #maximum of electronic step

EDIFF = 1E-5

LREAL = Auto

EDIFFG = -1E-2

NSW = 1000 # number of ionic step

IBRION = 2 # optimization

ISIF = 2

ISMEAR = 0

SIGMA = 0.2

IVDW=11

VDW_RADIUS=50.2

VDW_CNRRADIUS=20.0

VDW_S6=14.0

VDW_S8=16.0

VDW_SR=1.0

GGA=PE

末态结构优化

过渡态计算实例HER

初态结构优化后，得到能量和结构。



得到初态结构优化

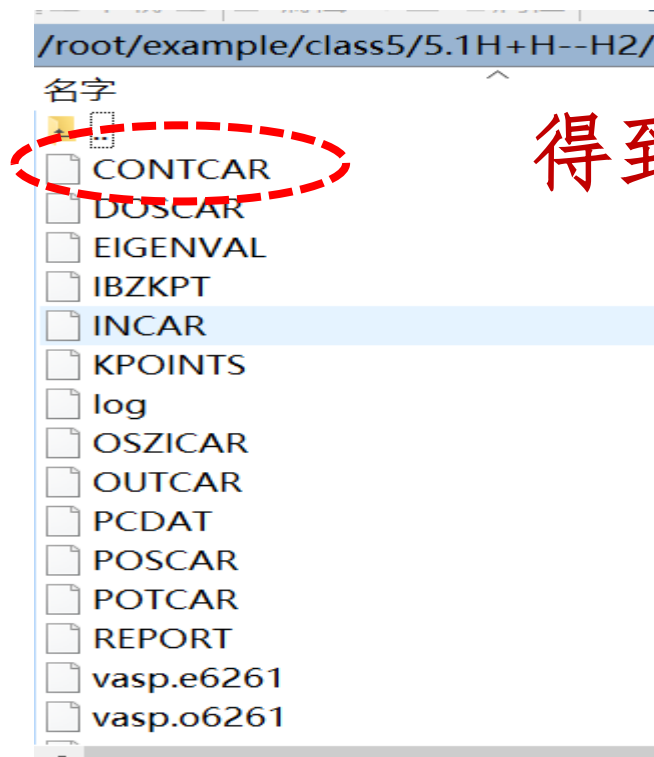
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

free energy TOTEN = -487.66046479 eV

energy without entropy= -487.61877707 energy(sigma->0) = -487.639620

得到初态结构能量

末态结构优化后，得到能量和结构。



得到末态结构优化

FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

free energy TOTEN = -487.73489353 eV

energy without entropy= -487.71341999 energy(sigma->0) = -487.724156

得到末态结构能量

中间点插点:

将得到的初态CONTCAR命名为**POSCAR-in**

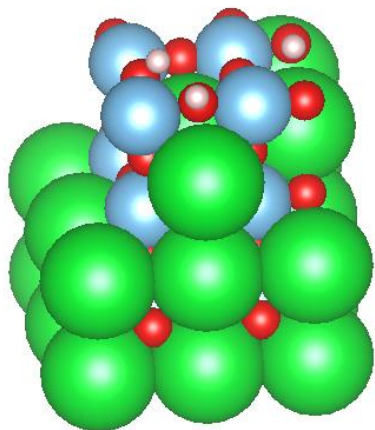
将得到的末态CONTCAR命名为**POSCAR-fin**

```
[root@iZuf6ewt8qjkgzelxd4lyr9Z test]# /usr/bin/vtstscripts-937/dist.pl POSCAR_in POSCAR_fin  
3.56816211012442
```

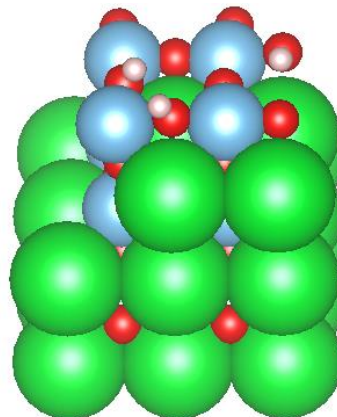
```
[root@iZuf6ewt8qjkgzelxd4lyr9Z test]# /usr/bin/vtstscripts-937/nebmake.pl POSCAR_in POSCAR_fin 4  
filetype1: vasp5  
filetype2: vasp5
```

```
OK, ALL SETUP HERE  
FOR LATER ANALYSIS, PUT OUTCARs IN FOLDERS 00 and 05 !!!
```

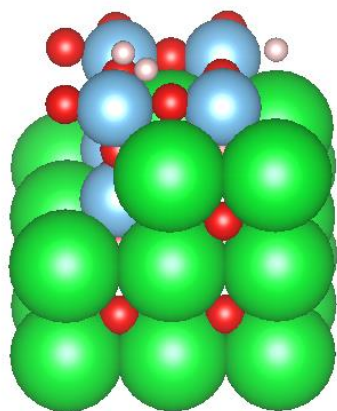
最后检查一下始末态和中间态的结构



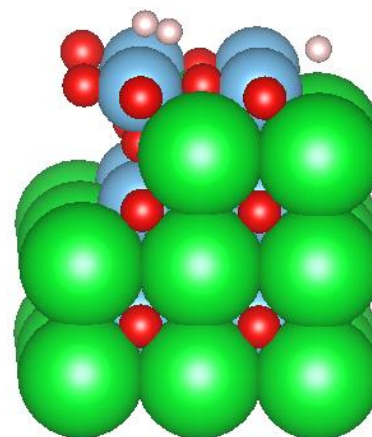
中间态 (01)



中间态 (02)



中间态 (03)



中间态 (04)

准备好上述文件就可以**提交过渡态**的计算具体操作如下：**qsub vasp.pbs**

名字



..



00



01



02



03



04



05



INCAR



KPOINTS



POTCAR



vasp.pbs

计算完成后查看能量：

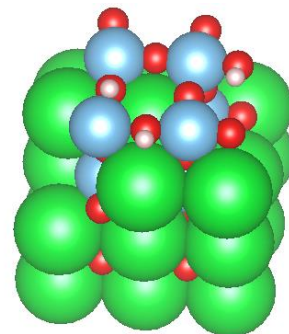
过渡态计算能量查看：输入**nebef.pl**

```
[root@iZuf6ewt8qjkzelxd4lyr9Z 5.1H+H--H2]# /usr/bin/vtstscripts-937/nebef.pl
```

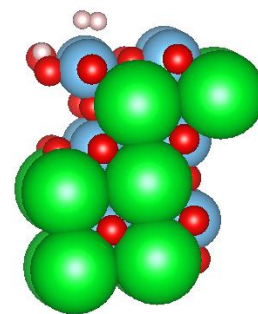
0	0.009618	-487.639600	0.000000	初态能量
1	0.023708	-487.229900	0.409700	
2	0.022362	-486.365000	1.274600	过渡态能量
3	0.028053	-487.433300	0.206300	
4	0.022266	-487.719500	-0.079900	
5	0.000000	-487.724100	-0.084500	末态能量

进行过渡态结构的导出和作图

过渡态能量: -486.37



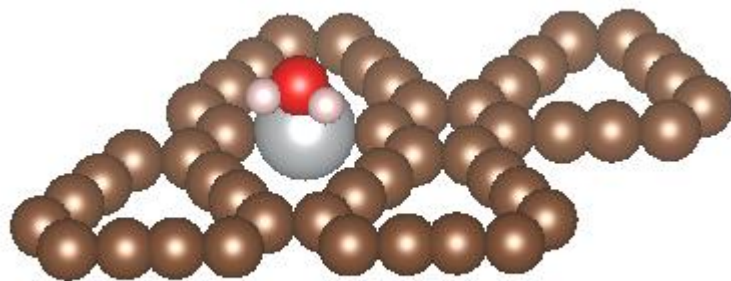
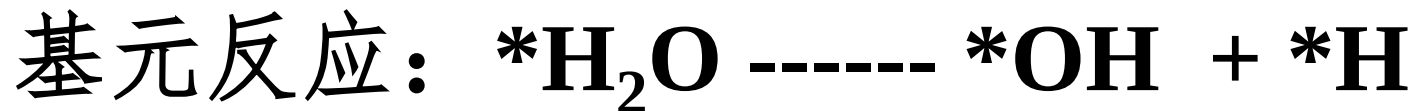
能垒 E_a : 1.27 eV



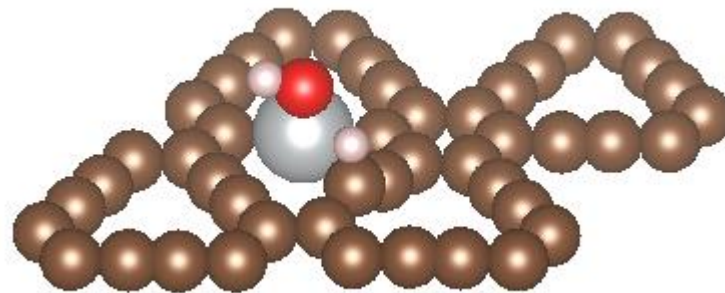
初态能量: -487.64 eV

反应能 E : -0.08 eV

末态能量: -487.72 eV



初态结构



末态结构

初态结构和末态结构优化，优化方法一样

/root/example/class5/test/IS

名字



..



INCAR



KPOINTS



POSCAR



POTCAR



vasp.pbs

Automatic mesh

0

G

1 1 1

0.0.0.

PREC = Normal

ISTART = 0

ICHARG = 2

ISPIN=2

ENCUT = 400

NELM = 500; #maximum of electronic step

EDIFF = 1E-5

LREAL = Auto

EDIFFG = -1E-2

NSW = 1000 # number of ionic step

IBRION = 2 # optimization

ISIF = 2

ISMEAR = 0

SIGMA = 0.2

IVDW=11

VDW_RADIUS=50.2

VDW_CNRRADIUS=20.0

VDW_S6=14.0

VDW_S8=16.0

VDW_SR=1.0

GGA=PE

初态结构优化

初态结构和末态结构优化，优化方法与上次课一样

/root/example/class5/test/FS

名字

..
INCAR
KPOINTS
POSCAR
POTCAR
vasp.pbs

末态结构优化

Automatic mesh

0

G

1 1 1

0.0.0.

PREC = Normal

ISTART = 0

ICHARG = 2

ISPIN=2

ENCUT = 400

NELM = 500; #maximum of electronic step

EDIFF = 1E-5

LREAL = Auto

EDIFFG = -1E-2

NSW = 1000 # number of ioinc step

IBRION = 2 # optimization

ISIF = 2

ISMEAR = 0

SIGMA = 0.2

IVDW=11

VDW_RADIUS=50.2

VDW_CNRADIUS=20.0

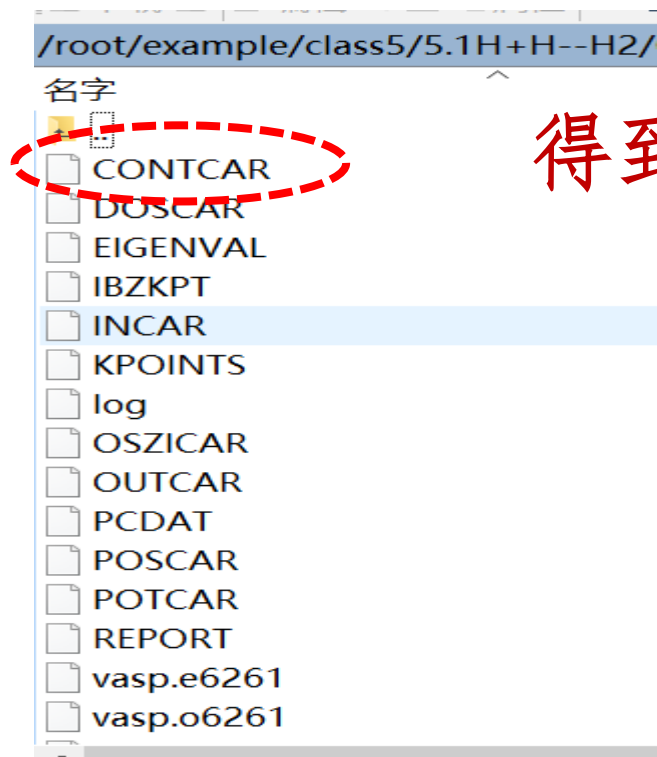
VDW_S6=14.0

VDW_S8=16.0

VDW_SR=1.0

GGA=PE

初态结构优化后，得到能量和结构。



得到初态结构优化

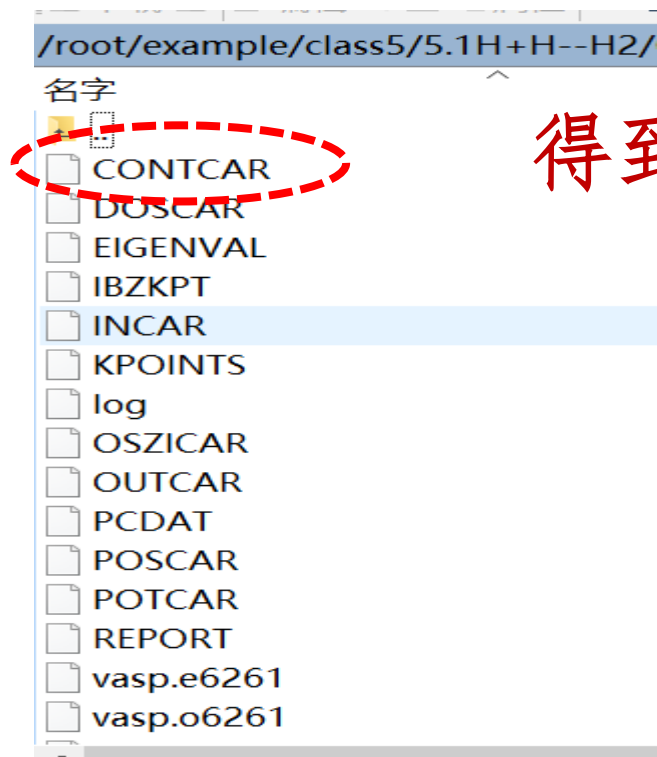
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

free energy TOTEN = -412.38789665 eV

energy without entropy= -412.10297873 energy(sigma->0) = -412.245437

得到初态结构能量

末态结构优化后，得到能量和结构。



得到末态结构优化

FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

free energy TOTEN = -411.60585531 eV

energy without entropy= -411.36930976 energy(sigma->0) = -411.487582

得到末态结构能量

中间点插点:

将得到的初态CONTCAR命名为**POSCAR-in**

将得到的末态CONTCAR命名为**POSCAR-fin**

```
/usr/bin/vtstscripts-937/dist.pl POSCAR_in
```

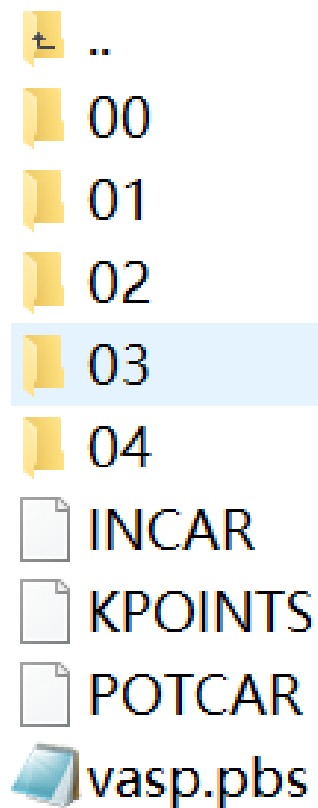
```
2.13880241372151  
-----
```

```
/usr/bin/vtstscripts-937/nebmake.pl POSCAR_in POSCAR_fin 3
```

```
filetype1: vasp5  
filetype2: vasp5
```

```
OK, ALL SETUP HERE  
FOR LATER ANALYSIS, PUT OUTCARs IN FOLDERS 00 and 04 !!!
```

准备好上述文件就可以**提交过渡态**的计算具体操作如下：**qsub vasp.pbs**



过渡态需要多次尝试！

问题一：中间的点能量比始末态还低！

```
[root@iZuf6ewt8qjkd41yr9Z 5.3H2O-H-OH]# nebef.pl
0          0.806755          -412.245400          0.000000
1          0.039534          -412.420400          -0.175000
2          0.030804          -411.217100          1.028300
3          0.041202          -411.416200          0.829200
```

可能原因：始末态不是最稳定的状态

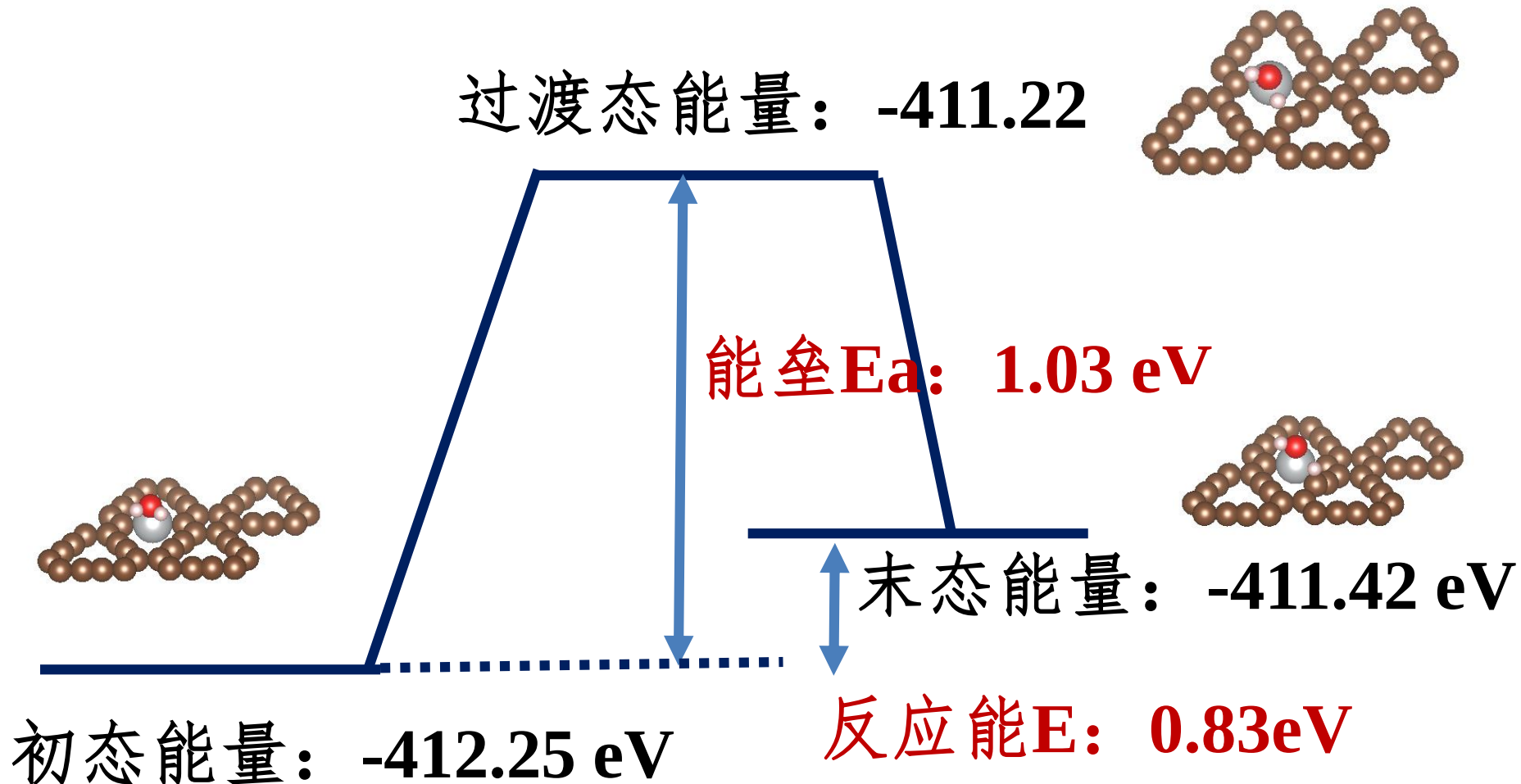
解决方案：重新优化始末态

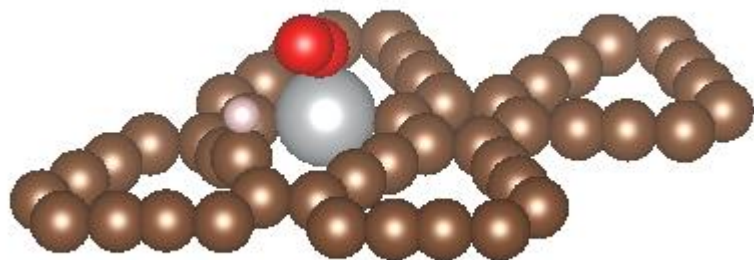
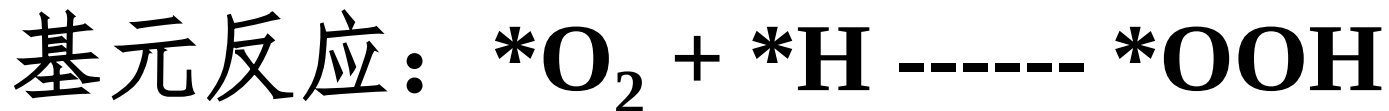
计算完成后查看能量：

过渡态计算能量查看：输入nebef.pl

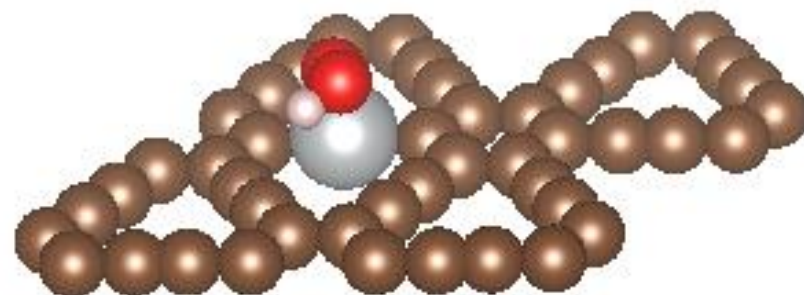
0	0.806755	-412.245400	0.000000	初态能量
1	0.039534	-412.420400	-0.175000	
2	0.030804	-411.217100	1.028300	过渡态能量
3	0.041202	-411.416200	0.829200	末态能量

进行过渡态结构的导出和作图





初态结构



末态结构

初态结构和末态结构优化，优化方法一样

/root/example/class5/test/IS

名字



..



INCAR



KPOINTS



POSCAR



POTCAR



vasp.pbs

Automatic mesh

0

G

1 1 1

0.0.0.

PREC = Normal

ISTART = 0

ICHARG = 2

ISPIN=2

ENCUT = 400

NELM = 500; #maximum of electronic step

EDIFF = 1E-5

LREAL = Auto

EDIFFG = -1E-2

NSW = 1000 # number of ionic step

IBRION = 2 # optimization

ISIF = 2

ISMEAR = 0

SIGMA = 0.2

IVDW=11

VDW_RADIUS=50.2

VDW_CNRRADIUS=20.0

VDW_S6=14.0

VDW_S8=16.0

VDW_SR=1.0

GGA=PE

初态结构优化

初态结构和末态结构优化，优化方法一样

/root/example/class5/test/FS

名字

..
INCAR
KPOINTS
POSCAR
POTCAR
vasp.pbs

末态结构优化

Automatic mesh

0

G

1 1 1

0.0.0.

PREC = Normal

ISTART = 0

ICHARG = 2

ISPIN=2

ENCUT = 400

NELM = 500; #maximum of electronic step

EDIFF = 1E-5

LREAL = Auto

EDIFFG = -1E-2

NSW = 1000 # number of ionic step

IBRION = 2 # optimization

ISIF = 2

ISMEAR = 0

SIGMA = 0.2

IVDW=11

VDW_RADIUS=50.2

VDW_CNRRADIUS=20.0

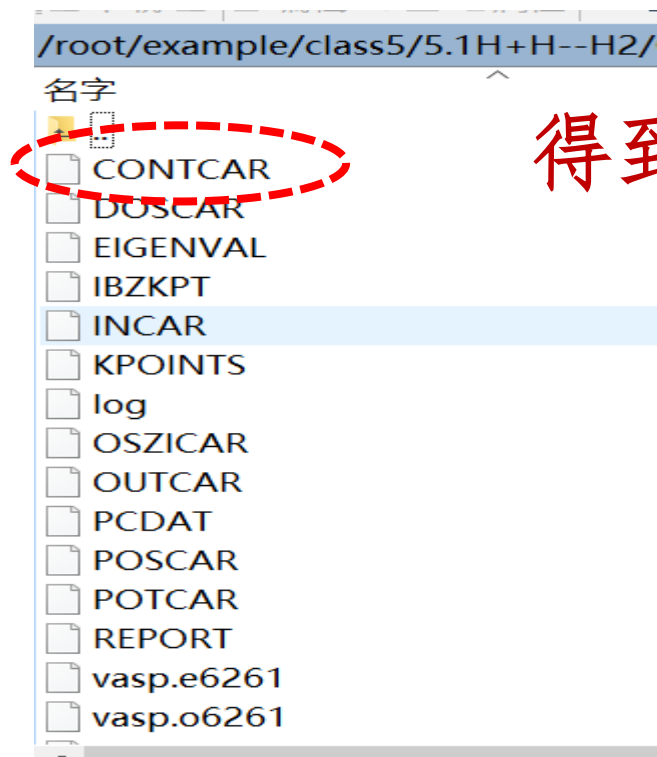
VDW_S6=14.0

VDW_S8=16.0

VDW_SR=1.0

GGA=PE

初态结构优化后，得到能量和结构。



得到初态结构优化

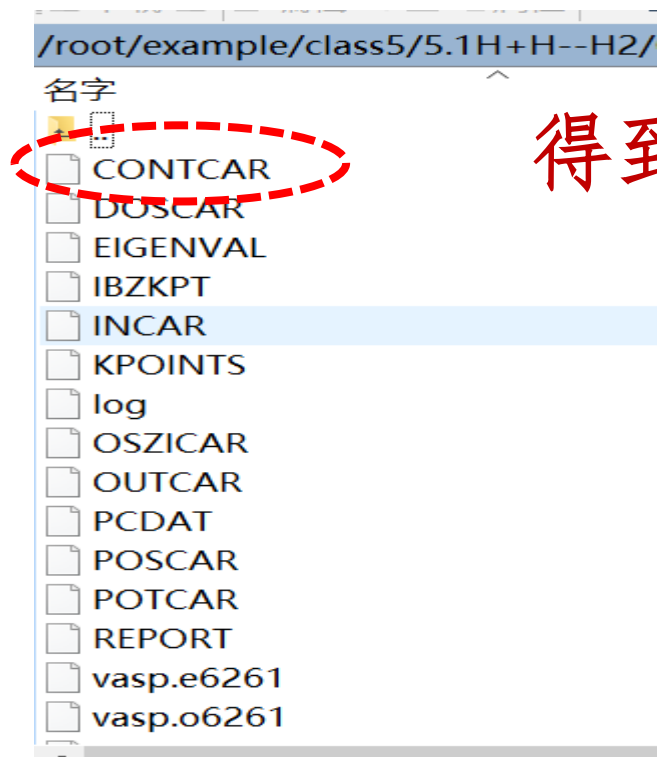
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

free energy TOTEN = -411.78622829 eV

energy without entropy= -411.44565490 energy(sigma->0) = -411.61594160

得到初态结构能量

末态结构优化后，得到能量和结构。



得到末态结构优化

FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)

free energy TOTEN = -412.52877575 eV

energy without entropy= -412.30391330 energy(sigma->0) = -412.416344

得到末态结构能量

中间点插点:

将得到的初态CONTCAR命名为**POSCAR-in**

将得到的末态CONTCAR命名为**POSCAR-fin**

```
/usr/bin/vtstscripts-937/dist.pl POSCAR_in
```

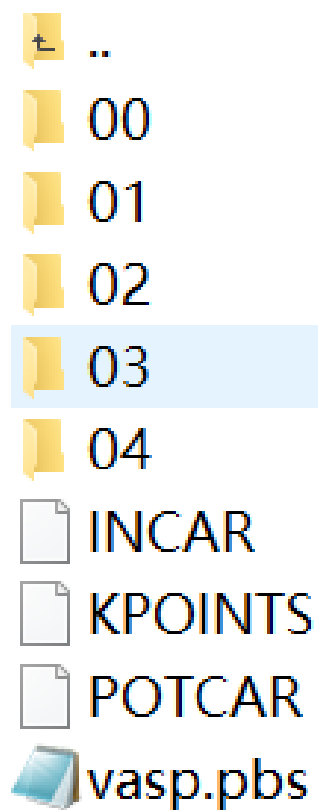
```
POSCAR_fin  
1.83550588219612
```

```
/usr/bin/vtstscripts-937/nebmake.pl POSCAR_in POSCAR_fin 3
```

```
filetype1: vasp5  
filetype2: vasp5
```

```
OK, ALL SETUP HERE  
FOR LATER ANALYSIS, PUT OUTCARs IN FOLDERS 00 and 04 !!!
```

准备好上述文件就可以**提交过渡态**的计算具体操作如下：**qsub vasp.pbs**



过渡态需要多次尝试！

问题一：中间的点能量比始末态还低！

```
[root@iZuf6ewt8qjkzelxd4lyr9Z TS]# nebef.pl
0          0.971473          -411.615900          0.000000
1          0.017303          -411.723900          -0.108000
2          0.049837          -411.628800          -0.012900
3          0.031453          -412.386800          -0.770900
```

可能原因：始末态不是最稳定的状态

解决方案：重新优化始末态

过渡态需要多次尝试！

问题二：能量一直震荡

593	F=	-.41034336E+03	E0=	-.41020417E+03	d E	=-.787517E-06
594	F=	-.41034336E+03	E0=	-.41020423E+03	d E	=-.377222E-05
595	F=	-.41034336E+03	E0=	-.41020404E+03	d E	=0.668571E-05
596	F=	-.41034336E+03	E0=	-.41020400E+03	d E	=-.506519E-06
597	F=	-.41034336E+03	E0=	-.41020398E+03	d E	=-.122494E-06
598	F=	-.41034336E+03	E0=	-.41020400E+03	d E	=0.141717E-06
599	F=	-.41034336E+03	E0=	-.41020401E+03	d E	=-.189175E-08
600	F=	-.41034336E+03	E0=	-.41020398E+03	d E	=-.825385E-07

可能原因：能量收敛精度不够；力不收敛；
优化算法问题；结构落入局部低点

解决方案：修改相应参数；微调原子原子

计算完成后查看能量：

过渡态计算能量查看：输入

/usr/bin/vtstscripts-937/nebef.pl

```
[root@iZuf6ewt8qjkzelxd4lyr9Z TS]# /usr/bin/vtstscripts-937/nebef.pl
```

0	0.032067	-411.771600	0.000000	初态能量
1	2.000000	-410.203900	1.567700	过渡态能量
2	50.000000	-411.098800	0.672800	
3	0.043959	-412.416300	-0.644700	末态能量

进行过渡态结构的导出和作图

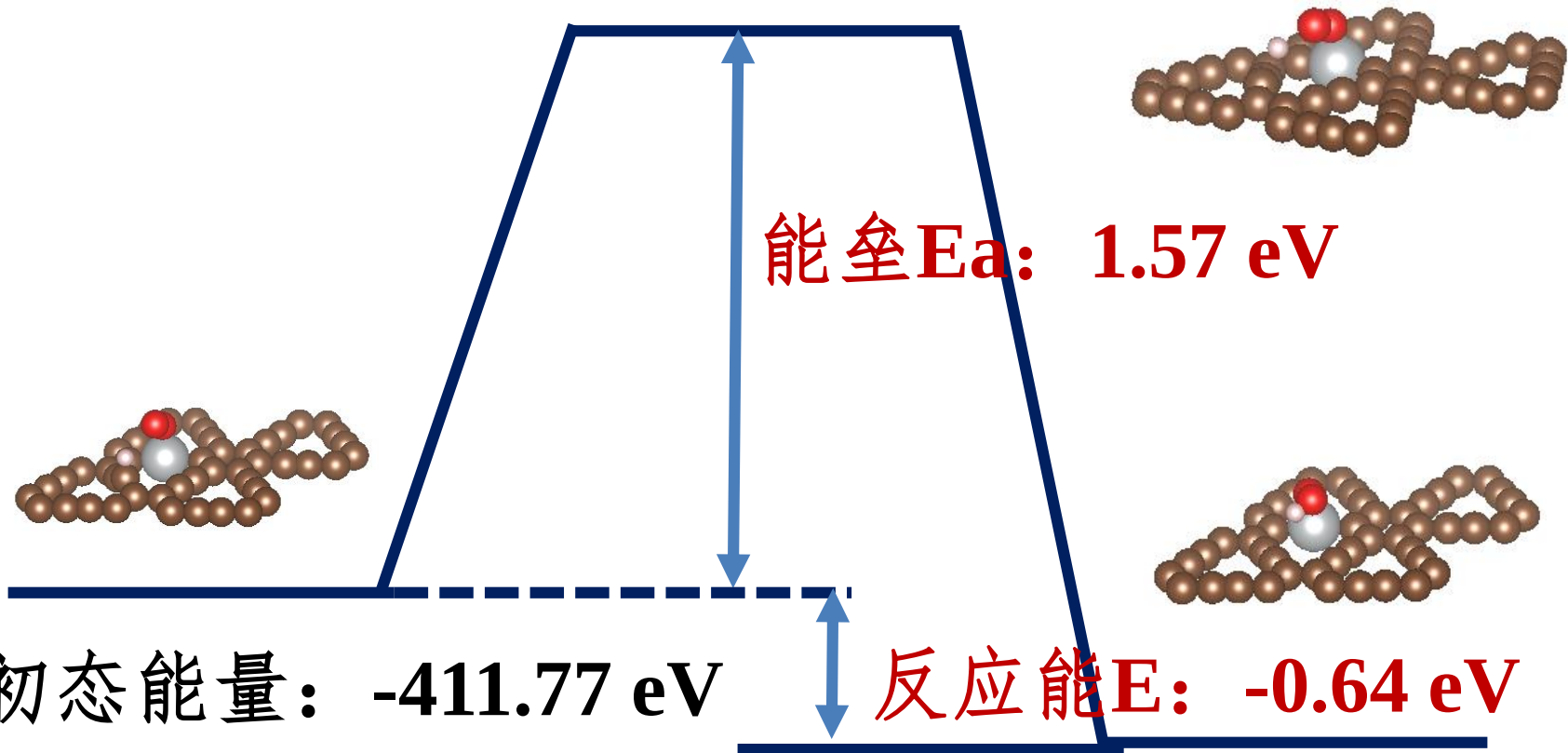
过渡态能量: -410.20

能垒 E_a : 1.57 eV

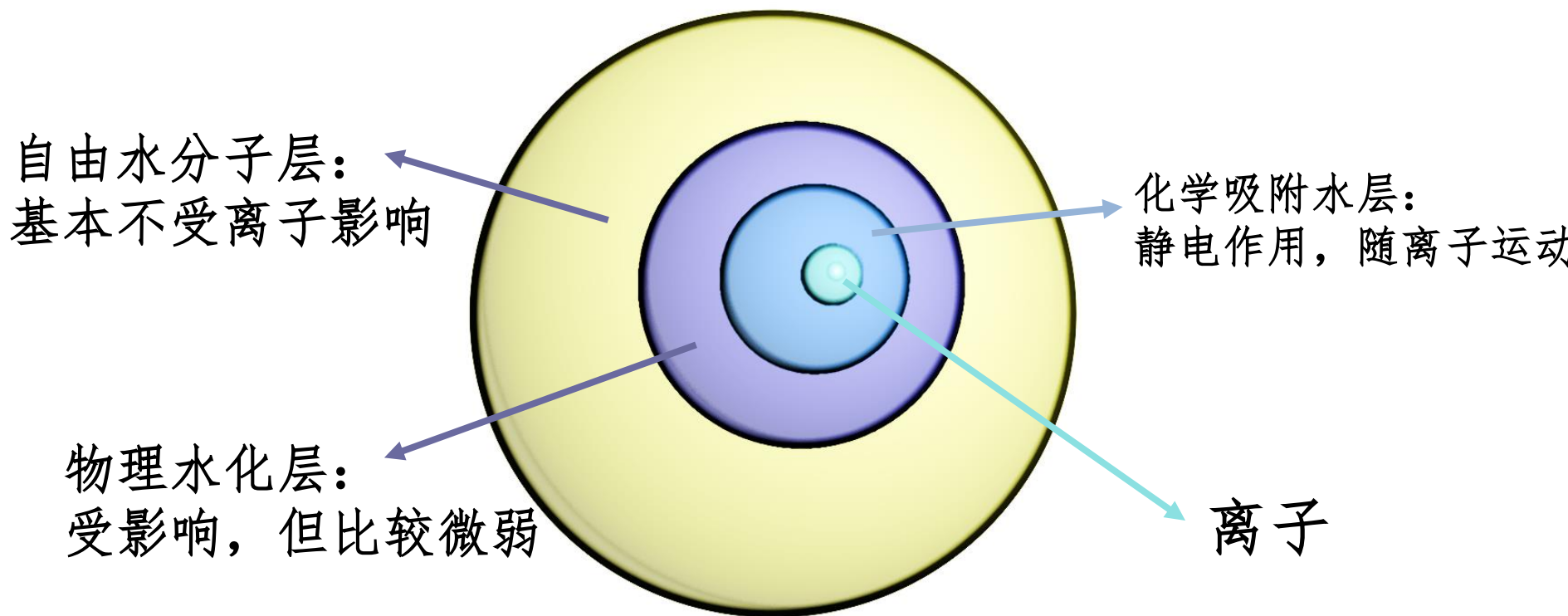
初态能量: -411.77 eV

反应能 E : -0.64 eV

末态能量: -412.41 eV



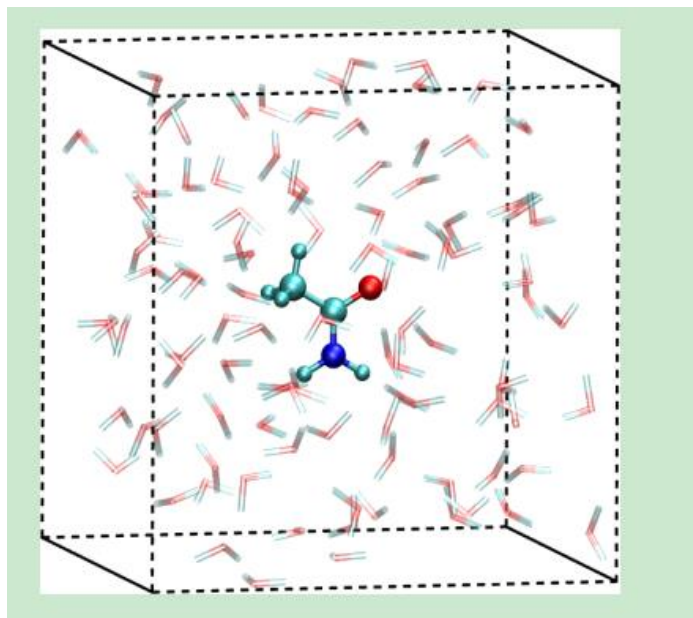
溶剂效应是指溶剂的物理和化学性质影响反应平衡和反应速度的效应。溶剂化本质主要是**静电作用**。



溶剂的存在将显著改变电子转移的动力学特征。

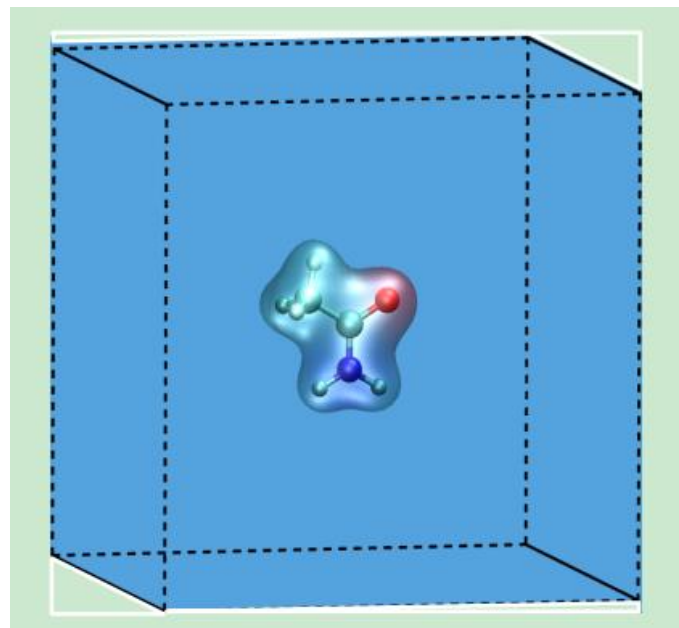
两大理论模型

显性溶剂模型



- ◆ 完全采用从头算的计算方法
- ◆ 详尽地展示计算系统的细节
- ◆ 需要研究溶剂分子的构型
- ◆ 计算耗费巨大

隐性溶剂模型



- ◆ 参数化的计算方法
- ◆ 将溶剂分子替换成连续介质
- ◆ 溶剂分子构型包含在参数中
- ◆ 计算相对廉价

隐性溶剂模型计算方法

HERBERT WERTHEIM COLLEGE OF ENGINEERING

UF

VASPsol Solid/Liquid Interfaces for DFT code VASP

VASPSOL

PUBLICATIONS

J. Chem. Phys. 140, 084106 (2014).
Phys. Rev. B 87, 245402 (2013).

VASPsol

VASPSOL

We implemented an implicit solvation model that describes the effect of electrostatics, cavitation, and dispersion on the interaction between a solute and solvent into the plane-wave DFT code VASP. Our implementation provides a computationally efficient means to calculate the effects of solvation on molecules and crystal surfaces as well as reaction barriers. The strength of our solvation model implementation is its capability to handle large periodic systems such as metal and semiconductor surfaces and its interoperability with standard ultrasoft pseudopotential and projector-augmented wave potential libraries. The software is freely available as a patch to the original VASP code.

For the latest version of code and documentation, please see our group's GitHub page:

<https://github.com/henniggroup/VASPsol>

Developers: Kiran Mathew and Richard G. Hennig

Collaborators: Ravishankar Sundararaman, Kendra Letchworth-Weaver, Tomas A. Arias

© 2014 Richard Hennig

隐性溶剂模型计算方法

requests Actions Projects Wiki Security Insights

master 3 branches 1 tag Go to file Code

ericfonseca95 Update README.md 0dc6b89 on 14 Apr 94 commits

docs	使用手册	Update USAGE.md	3 years ago
examples	算例	reorganized the files, modules in their files in the src/modules folder	6 years ago
src	solvation.F 文件	Add files via upload	6 months ago
.gitignore		add pbz patch for version 5.4.1. update readme	5 years ago
LICENSE.rst		add license file	6 years ago
README.md	安装教程	Update README.md	6 months ago

隐性溶剂模型计算方法

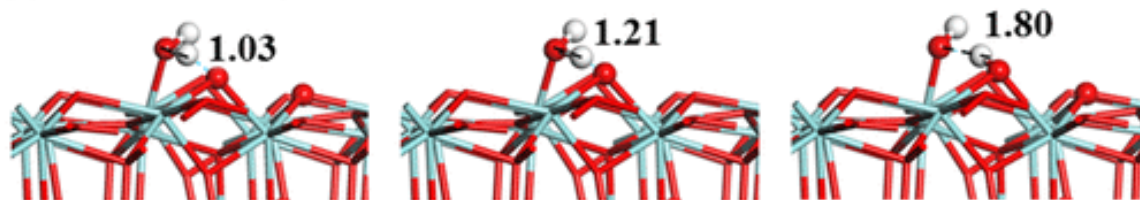
1. 真空条件下的结构优化，并输出WAVECAR用于下一步计算
2. 隐形溶剂模型计算INCAR，最简单的情况只需要设置参数
LSOL = .TRUE.

```
1 SYSTEM = CO
2 ISTART = 1
3 ENCUT = 800
4 PREC = Accurate
5 ISMEAR = 0
6 SIGMA = 0.01
7 EDIFF = 1E-6
8
9 LSOL = .TRUE.
10 EB_K = 80
```

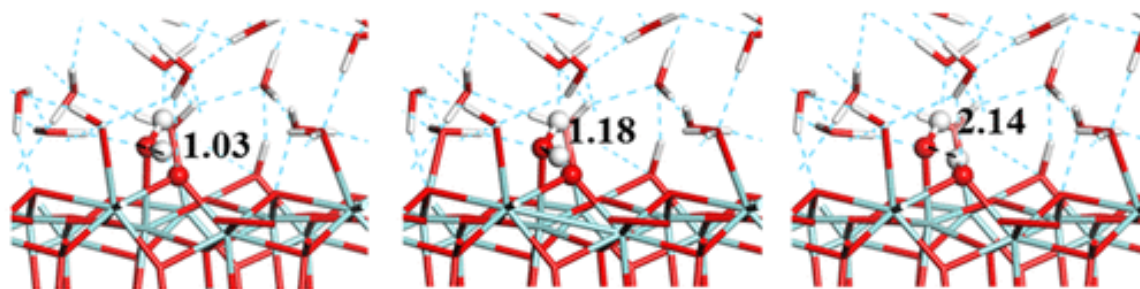
3. 优化后的CONTCAR另存为POSCAR，加上WAVECAR，其与输入文件不变。

显性溶剂模型计算方法

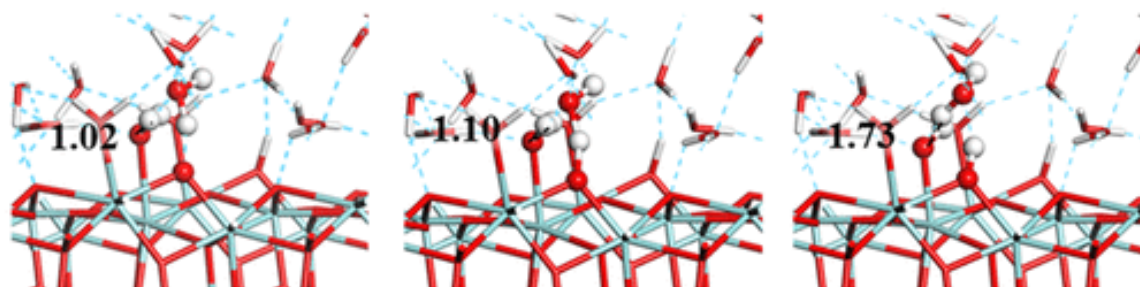
(a) $\Delta E^\ddagger = 8 \text{ kJ/mol}$; $\Delta E = -9 \text{ kJ/mol}$



(b) $\Delta E^\ddagger = 7 \text{ kJ/mol}$; $\Delta E = -31 \text{ kJ/mol}$



(c) $\Delta E^\ddagger = 7 \text{ kJ/mol}$; $\Delta E = -33 \text{ kJ/mol}$



显性溶剂模型计算方法

一般先跑一个AIMD让溶剂分子模型趋于收敛。

```
1 SYSTEM = graphyne opt
2 PREC = Normal
3
4 #start running#
5 ISTART = 0
6 ICHARG = 2
7 ISPIN = 2
8 #MAGMOM =
9 LWAVE = .FALSE
10 LCHARG = .FALSE.
11 #LELF = .TRUE.
12 LREAL = Auto
13
14 #electronic opt SCF#
15 ENCUT = 400
16 NELM = 500
17 #NELMIN = 6
18 EDIFF = 1E-5
19 GGA = PE
20 ALGO = Fast
21 IALGO=48
22
23 #ino opt SCF#
24 EDIFFG = -0.02
25 ISIF = 2
26 ISYM = 0
27
28 #other#
29 ISMEAR = 0
30 SIGMA = 0.1
31 #IVDW = 12
32 NCORE = 4
33
34 ##MD
35
36 TEBEG=500
37 TEEND=500
38 NSW=600
39 IBRION=0
40 POTIM=0.5
41 SMASS=0
42 NBLOCK= 1
43
```

显性溶剂模型计算方法

ALGO = Normal | VeryFast | Fast | All | Damped

Default:

ALGO = Normal

The ALGO tag is a convenient way to specify the electronic minimisation algorithm in VASP.4.5 and later versions.

- ALGO = Normal, IALGO=38 (blocked Davidson block iteration scheme)
- ALGO = Very Fast, IALGO=48 (RMM-DIIS)
- ALGO = Fast, 先是IALGO=38 , 然后IALGO=48
- ALGO = All, IALGO=58, 波函数所有的能带同时更新
- ALGO = Damped , IALGO=53 (damped velocity friction algorithm)

注意：计算大体系AIMD计算时，ALGO=Very Fast的话，需设置LREAL=A来加速计算，至少设置NELMIN=4。

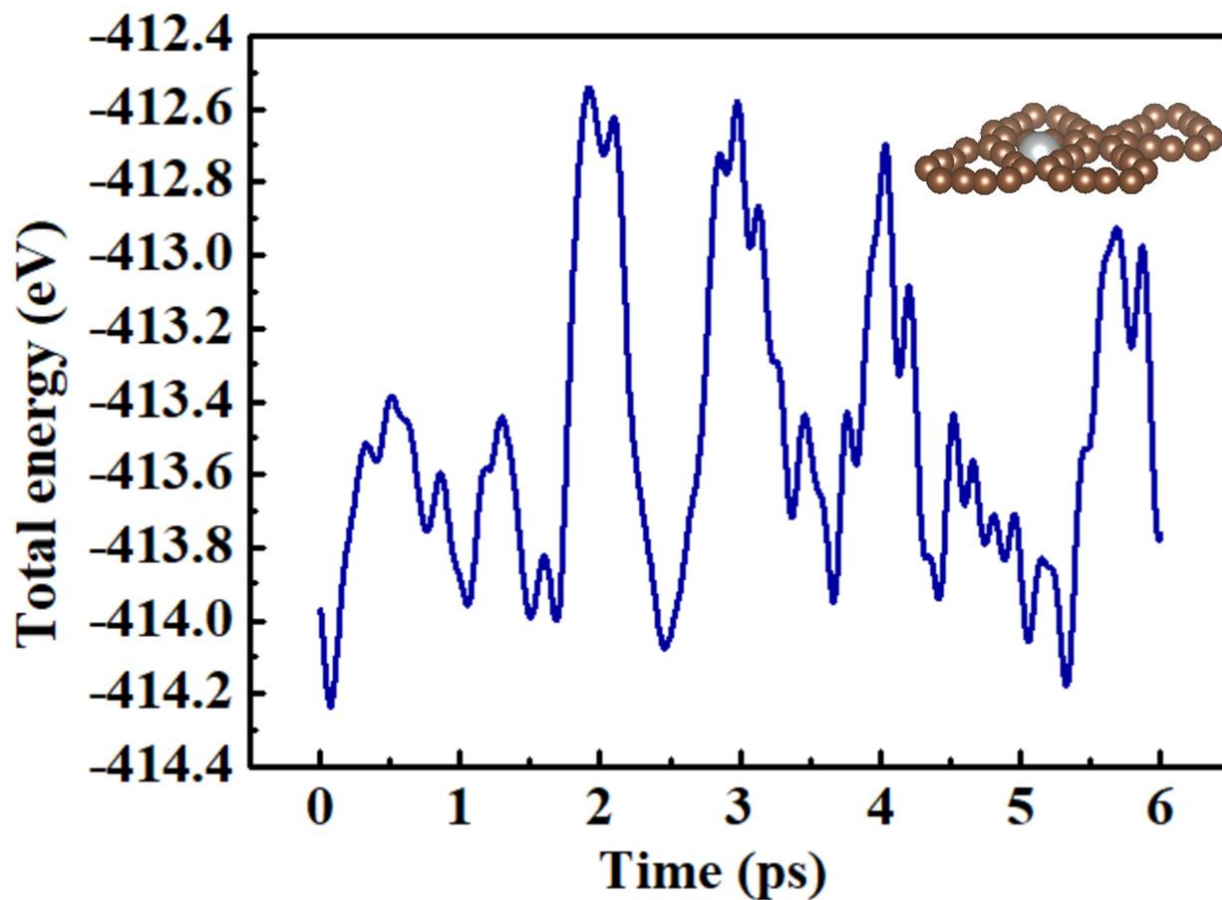
显性溶剂模型计算方法

一般先跑一个AIMD让溶剂分子模型趋于收敛。

```
1 SYSTEM = graphyne opt
2 PREC = Normal
3
4 #start running#
5 ISTART = 0
6 ICHARG = 2
7 ISPIN = 2
8 #MAGMOM =
9 LWAVE = .FALSE
10 LCHARG = .FALSE.
11 #LELF = .TRUE.
12 LREAL = Auto
13
14 #electronic opt SCF#
15 ENCUT = 400
16 NELM = 500
17 #NELMIN = 6
18 EDIFF = 1E-5
19 GGA = PE
20 ALGO = Fast
21 IALGO=48
```

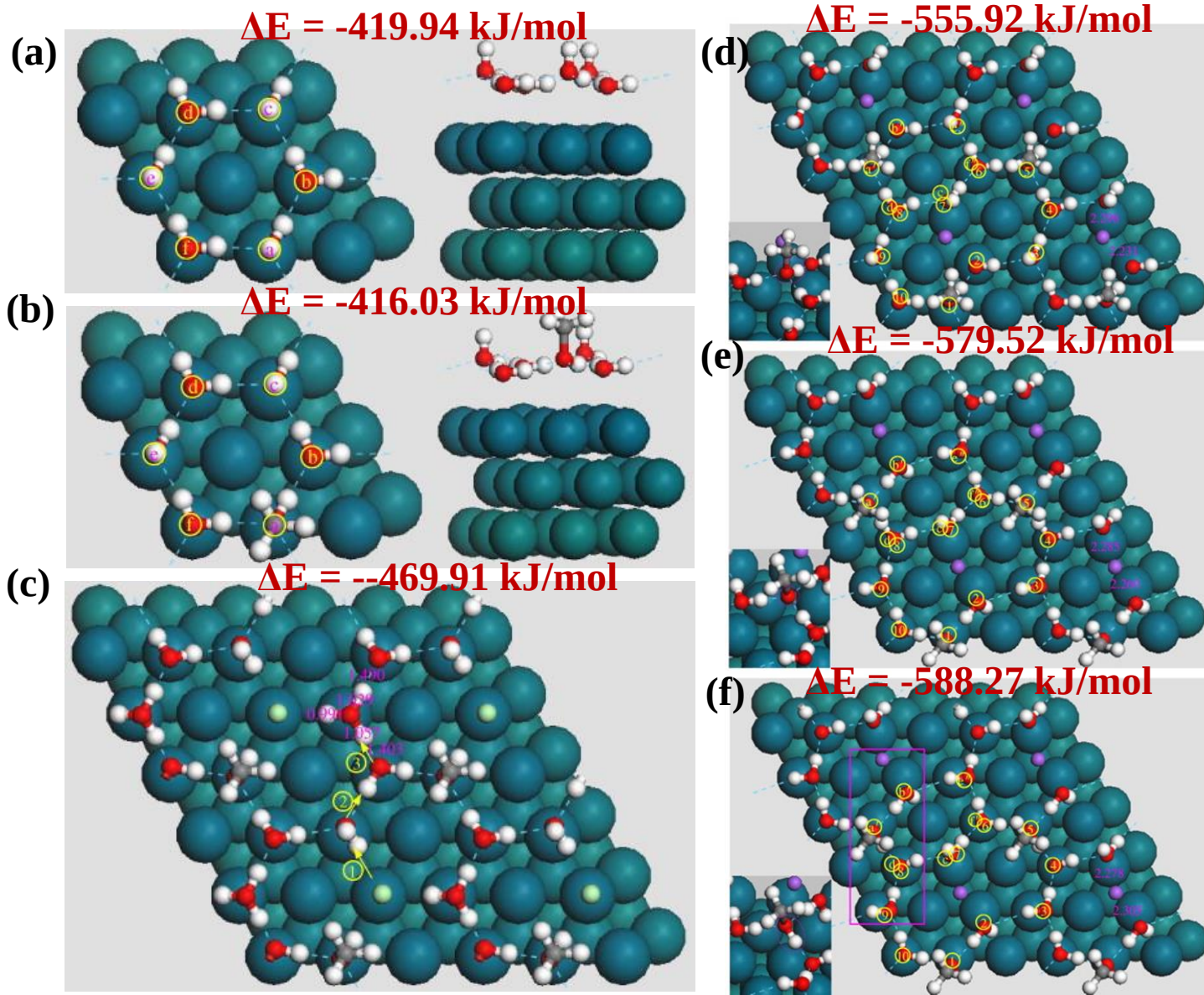
```
23 #ino opt SCF#
24 EDIFFG = -0.02
25 ISIF = 2
26 ISYM = 0
27
28 #other#
29 ISMEAR = 0
30 SIGMA = 0.1
31 #IVDW = 12
32 NCORE = 4
33
34 ##MD
35
36 TEBEG=500      起始温度
37 TEEND=500     结束温度
38 NSW=600       步数
39 IBRION=0      开启MD计算
40 POTIM=0.5     步长
41 SMASS=0       控制MD的速度
42 NBLOCK= 1
43
```

显性溶剂模型计算结果 (AIMD)

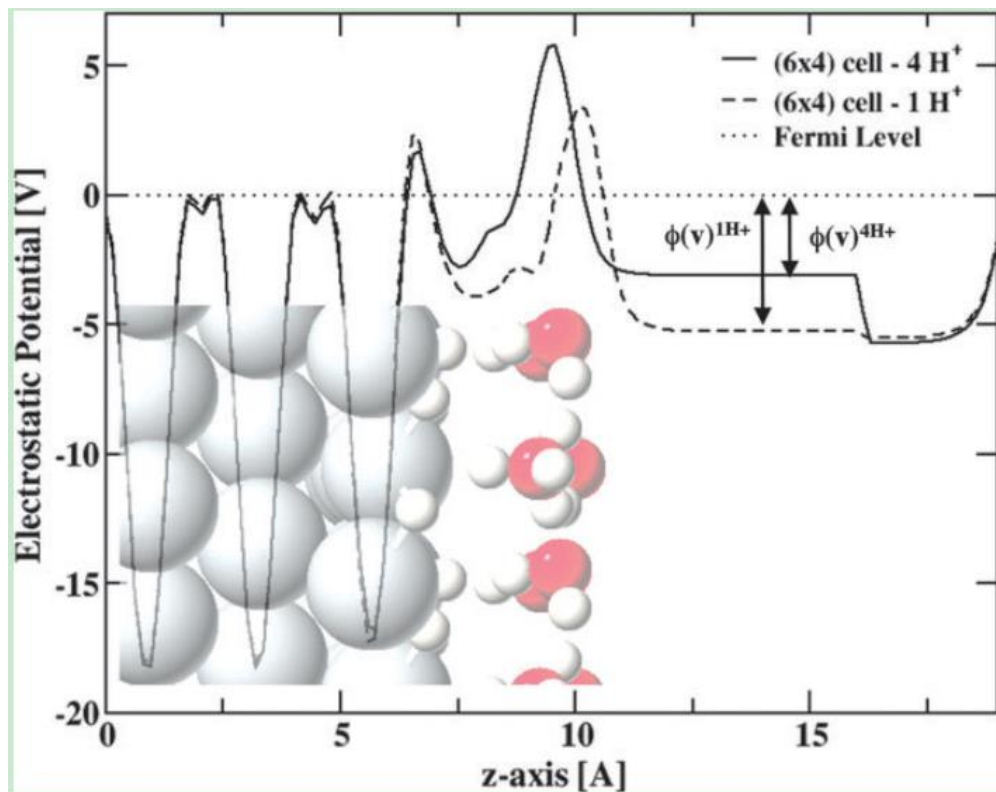


显性溶剂模型计算实例

Journal of Molecular Structure 980 (2010) 208–213



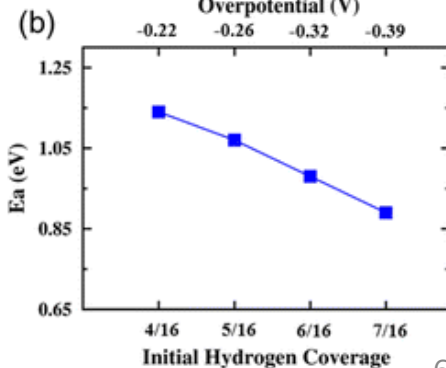
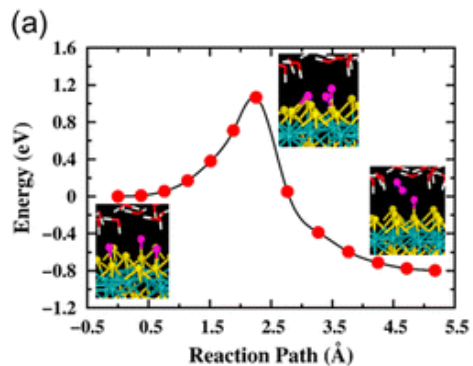
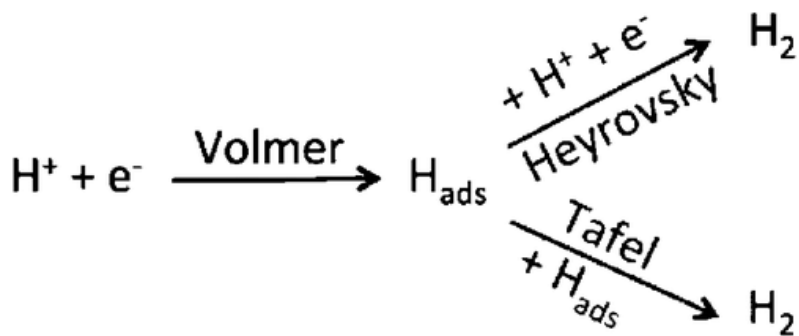
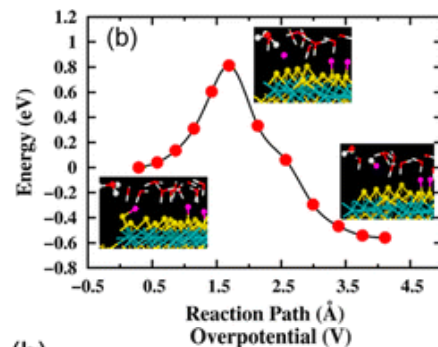
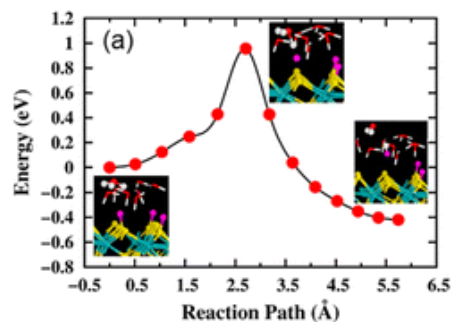
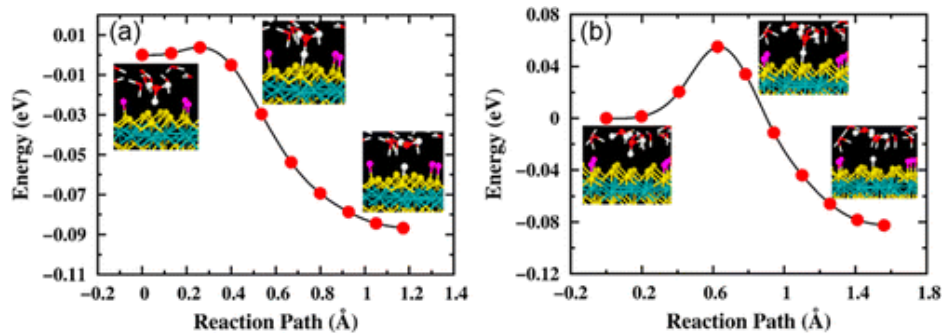
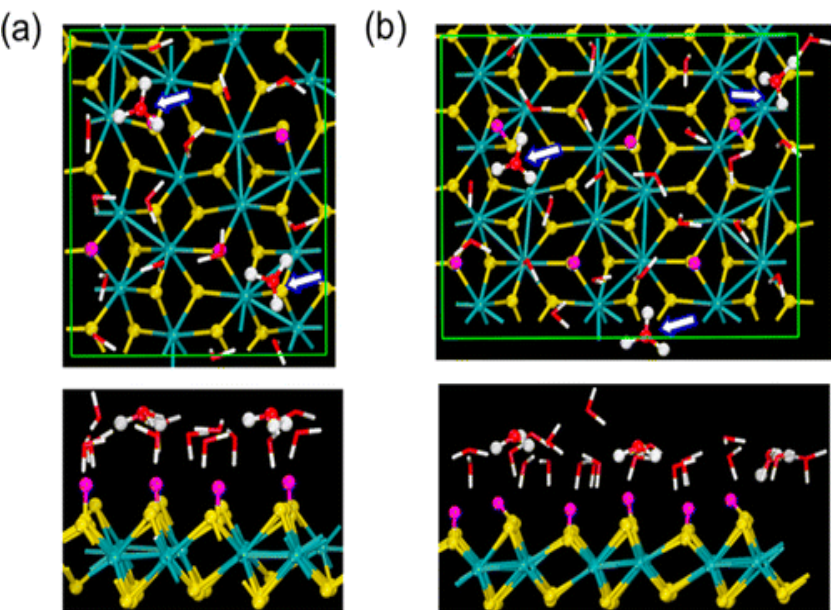
Phys. Chem. Chem. Phys., 2007, 9, 3241–3250



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显性溶剂模型计算实例

ACS Catal. 2016, 6, 8, 4953–4961



Thank you !